Chemistry Topic 6 — Gases, Fuels, Energy and Combustion

Sources baked in: NCERT Class 8, 10 & 11 (Combustion and Flame; Sources of Energy; States of Matter: Gases), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — High-frequency recurring questions on LPG constituents (Butane + Propane), Ethyl mercaptan odorant, CNG vs Biogas primary gas (Methane), Water gas ($CO + H_2$) vs Producer gas ($CO + N_2$), Coal ranks by carbon content, and highest calorific value (Hydrogen). **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**National Green Hydrogen Mission**	Targeted green hydrogen production capacity of $5\ \text{MMT}$ per annum by 2030. Hydrogen (H_2) possesses the **highest gravimetric calorific value of all fuels ($150,000\ \text{kJ/kg}$)** and burns to produce pure water ($2H_2 + O_2 \rightarrow 2H_2O$) with zero greenhouse gas emissions.
2025	**UPPCS Prelims 2025**	Direct question on **Biogas Major Constituents**: Keyed strictly as **Methane (CH_4, 55–70%) + Carbon Dioxide (CO_2, 30–45%)**. Distractors like Butane (LPG) and Carbon Monoxide (Water/Producer gas) were eliminated.
2024–2025	**E20 Ethanol Blending Target**	India successfully rolled out **E20 Fuel** (20% bio-ethanol blended with 80% motor gasoline/petrol) nationwide across retail fuel pumps, advancing the target date from 2030 to 2025. Ethanol acts as a high-octane oxygenate reducing carbon monoxide tailpipe emissions.
2022	**UPPCS Prelims 2022**	Re-tested identity of **CNG** and **Natural Gas**: Both consist predominantly of **Methane (CH_4, $> 85\%$)**. Mechanical compression to $200\ \text{bar}$ does not alter the fundamental chemical identity of the gas.
2021	**UPPCS Prelims 2021**	Tested **LPG Primary Constituent**: Correctly keyed as **Butane** (together with propane). Highlighted the role of **Ethyl Mercaptan (C_2H_5SH)** as the vital foul-smelling leak indicator added to odorless LPG cylinders.

Detailed Theory Notes

6.1 Fuel Thermodynamics & Calorific Value Hierarchy

1. Defining Principles

- The chemical energy stored within covalent bonds of fossil fuels and biofuels is released as thermal kinetic energy when carbon and hydrogen atoms oxidize into CO_2 and H_2O:

$$C + O_2 \rightarrow CO_2 + 393.5\ \text{kJ/mol}$$

$$2H_2 + O_2 \rightarrow 2H_2O + 571.6\ \text{kJ/mol}$$

- **Higher Calorific Value (Gross Calorific Value, GCV)**: Total heat released when unit mass of fuel burns completely and combustion products (H_2O) are cooled back to standard reference temperature (25°C), condensing steam and capturing its latent heat of vaporization (L_v).
- **Lower Calorific Value (Net Calorific Value, NCV)**: Effective heat liberated when steam produced during combustion escapes into the atmosphere without condensing:

$$\text{NCV} = \text{GCV} - (\text{Mass of water formed} \times L_v)$$

- **Hydrogen's Superiority**: Gravimetrically, hydrogen possesses the highest energy density of any chemical fuel ($150\ \text{MJ/kg}$), because hydrogen has the lowest atomic mass and highest charge-to-mass ratio among combustible elements.

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Gravimetric Calorific Value Sequence:
Hydrogen (150,000 kJ/kg) >> Methane (55,000) > LPG/CNG (50,000) > Petrol/Diesel (45,000) > Coa
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6.2 Gaseous Fuels: Petroleum Gas, Natural Gas & Biogas

1. Liquefied Petroleum Gas (LPG)

- **Thermodynamic Transition**: At atmospheric pressure ($1\ \text{atm}$), butane boils at -0.5°C and propane boils at -42°C. When compressed inside a steel cylinder at $6\text{–}8\ \text{bar}$ ($0.6\text{–}0.8\ \text{MPa}$), these gases condense into liquid form at ambient room temperature.
- **Vapor-Liquid Equilibrium**: Inside the cylinder, liquid LPG sits at the bottom with pressurized gas filling the headspace above. When the valve is opened, pressure drops, causing the liquid to boil instantly into gas for combustion.
- **Safety Engineering — Ethyl Mercaptan**: Because hydrocarbons are odorless, any minor valve leak would be undetectable, accumulating near the floor (density of butane vapor ≈ 2.0 relative to air $= 1.0$). Blending **Ethyl Mercaptan (C_2H_5SH)** ensures human olfactory detection at concentrations as low as $1\ \text{part per billion (ppb)}$, well below the lower explosive limit (1.8%).

2. Biogas / Gobar Gas Chemistry

- **Anaerobic Biodigestion Stages**: 1. *Hydrolysis*: Complex biopolymers (cellulose, hemicellulose, proteins) are broken down by extracellular bacterial enzymes into soluble monomers (glucose, amino acids). 2. *Acidogenesis*: Acid-forming bacteria convert monomers into volatile fatty acids (acetic acid, propionic acid, butyric acid), along with CO_2 and H_2. 3. *Acetogenesis*: Acetogenic bacteria convert fatty acids into acetic acid (CH_3COOH), CO_2, and H_2. 4. *Methanogenesis*: Strictly anaerobic methanogenic archaebacteria (*Methanobacterium*, *Methanococcus*) convert acetate and hydrogen into methane:

$$CH_3COOH \rightarrow CH_4 + CO_2$$

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

- The overall biogas output contains $55\text{–}70\%\ \text{Methane}\ (CH_4)$ and $30\text{–}45\%\ \text{Carbon Dioxide}\ (CO_2)$.

6.3 Industrial Fuel Gases: Water Gas, Producer Gas & Syngas

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                      ┌── Water Gas (Syngas): Coke + Steam (1000°C) ──> CO + H₂ (Endothermic)
Industrial Gas Mixes ─┤
                      └── Producer Gas:       Coke + Air (1100°C)   ──> CO + N₂ (Exothermic)
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1. Water Gas (Syngas)

- Formed by passing steam through a bed of incandescent red-hot coke at temperatures above 1000°C:

$$C\,(\text{s}) + H_2O\,(\text{g}) \xrightarrow{1000^\circ\text{C}} CO\,(\text{g}) + H_2\,(\text{g}) \quad (\Delta H = +131.3\ \text{kJ/mol})$$

- Because the reaction is strongly endothermic, the coke bed cools down rapidly below the reaction threshold. To maintain production, an alternating process is used: an "air blow" (burning coke exothermically with air to reheat the bed) alternated with a "steam run" (injecting steam to generate water gas).
- Used for synthesizing industrial methanol ($CO + 2H_2 \xrightarrow{ZnO/Cr_2O_3} CH_3OH$) and as a source of commercial hydrogen via the **Water-Gas Shift Reaction**:

$$CO + H_2O\,(\text{steam}) \xrightarrow{Fe_2O_3/Cr_2O_3,\,450^\circ\text{C}} CO_2 + H_2$$

2. Producer Gas

- Manufactured by blowing air through a thick bed of red-hot coke in a furnace called a gas producer at 1100°C:

$$2C\,(\text{s}) + O_2\,(\text{g}) + 4N_2\,(\text{g}) \rightarrow 2CO\,(\text{g}) + 4N_2\,(\text{g}) \quad (\Delta H = -221\ \text{kJ/mol})$$

- The reaction is exothermic, keeping the furnace continuously hot without auxiliary heating.
- Because it contains roughly 60% non-combustible atmospheric nitrogen, it burns with a low flame temperature, but is economical for firing large metallurgical furnaces.

6.4 Solid & Liquid Fuels: Petroleum Refining & Combustion Quality

1. Coal Metamorphism (Coalification)

Vegetation buried under ancient swamp sediments underwent slow biochemical decay by anaerobic bacteria, followed by geological compression under immense overburden pressure and geothermal heat over hundreds of millions of years:

$$\text{Wood} \xrightarrow{\text{bacterial decay}} \text{Peat} \xrightarrow{\Delta, P} \text{Lignite} \xrightarrow{\Delta, P} \text{Bituminous Coal} \xrightarrow{\Delta, P} \text{Anthracite Coal}$$

With increasing metamorphic rank, carbon percentage increases ($< 60\% \rightarrow > 90\%$), volatile matter and moisture drop, and calorific value climbs from $10\ \text{MJ/kg}$ to $33\ \text{MJ/kg}$.

2. Octane vs. Cetane Engine Physics

- **Octane Rating (Spark-Ignition / Petrol Engines)**:

- Petrol engines compress a premixed air-fuel vapor. If the fuel detonates prematurely before the timed spark occurs, knocking occurs.
- Straight-chain alkanes (like n-heptane, $CH_3 - (CH_2)_5 - CH_3$) form free radicals easily and knock severely (Octane rating = 0).
- Highly branched alkanes (like iso-octane / 2,2,4-trimethylpentane) and aromatics resist pre-ignition (Octane rating = 100).
- **Cetane Rating (Compression-Ignition / Diesel Engines)**:
- Diesel engines compress air alone to high pressures ($30–40\ \text{bar}$), raising its temperature to $\approx 600^\circ\text{C}$. Diesel fuel is then sprayed directly into the hot air and must auto-ignite spontaneously with minimal delay.
- Straight-chain alkanes (like cetane / n-hexadecane, $C_{16}H_{34}$) auto-ignite rapidly (Cetane rating = 100).
- Highly branched hydrocarbons take too long to auto-ignite, resulting in diesel knocking.

6.5 Biofuels, Ethanol Blending & Clean Energy Transitions

1. The National Biofuel Policy & E20 Integration

- Under India's updated National Policy on Biofuels, the target for 20% **ethanol blending in petrol (E20)** was achieved ahead of schedule by 2025–2026.
- **Technical Mechanism**:
- Ethanol (CH_3CH_2OH) has a high blending octane rating ($\approx 108–110$), boosting the octane number of standard motor gasoline without toxic metallic additives.
- As an oxygenated fuel ($34.7\%\ \text{oxygen}$), ethanol promotes complete hydrocarbon combustion in engine cylinders, reducing tailpipe emissions of unburnt hydrocarbons and carbon monoxide (CO).
- **2G Ethanol Production**: Lignocellulosic agricultural residues (paddy straw, wheat straw) are pretreated with enzymes to break down cellulose and hemicellulose into fermentable pentose and hexose sugars, which are fermented into fuel-grade anhydrous ethanol (99.5% purity).

2. Biodiesel from Non-Edible Seeds (Jatropha & Karanja)

- Inedible seed oils from *Jatropha curcas* and *Pongamia pinnata* contain high molecular weight triglycerides that cannot be burned directly in diesel engines due to high viscosity ($10\times$ diesel) and carbon deposition.
- **Transesterification Process**:

$$\text{Triglyceride} + 3CH_3OH\ (\text{Methanol}) \xrightarrow{NaOH/KOH} 3\ \text{FAME (Biodiesel)} + \underset{\text{Byproduct}}{\text{Glycerol}}$$

The resulting Fatty Acid Methyl Esters (FAME) have physical viscosity, cetane rating, and combustion properties nearly identical to petroleum diesel.

Bilingual Terminology -- Gases, Fuels and Energy Chemistry

English Term	Hindi Term	Key Anchor
Combustion (दहन)	दहन	Chemical reaction of fuel with O2 producing heat and light
Fuel (ईंधन)	ईंधन	Substance that releases energy by combustion
Calorific Value (ऊष्मीय मान)	ऊष्मीय मान	Energy released per kg of fuel on complete combustion; kJ/kg
LPG (द्रवित पेट्रोलियम गैस)	द्रवीभूत पेट्रोलियम गैस	Liquefied Petroleum Gas; propane + butane; ethyl mercaptan (smell)
CNG (संपीडित प्राकृतिक गैस)	संपीडित प्राकृतिक गैस	Compressed Natural Gas; mainly methane; cleaner fuel
Coal (कोयला)	कोयला	Fossil fuel; types by carbon content: Peat < Lignite < Bituminous < Anthracite
Petroleum (पेट्रोलियम)	पेट्रोलियम	Crude oil; mixture of hydrocarbons; refined by fractional distillation
Producer Gas (प्रोड्यूसर गैस)	प्रोड्यूसर गैस	CO + N2; made by passing air over hot coke; fuel gas
Water Gas (जल गैस)	जल गैस	CO + H2; made by passing steam over hot coke; synthesis gas
Biogas (बायोगैस)	बायोगैस	CH4 (55-75%) + CO2; from anaerobic decomposition; clean fuel
Octane Number (ऑक्टेन संख्या)	ऑक्टेन संख्या	Measures anti-knock quality of petrol; higher = better; isooctane = 100
Photosynthesis (प्रकाश संश्लेषण)	प्रकाश-संश्लेषण	CO2 + H2O + light energy -> glucose + O2; solar energy storage
Ozone (ओजोन)	ओजोन	O3; protects Earth from UV; ozone layer at 15-35 km altitude
Greenhouse Gas (ग्रीनहाउस गैस)	हरित गृह गैस	CO2, CH4, N2O, CFCs; traps heat; causes global warming
Renewable Energy (नवीकरणीय ऊर्जा)	नवीकरणीय ऊर्जा	Solar, wind, hydro, geothermal, tidal; replenishable
Non-renewable Energy (अनवीकरणीय ऊर्जा)	अनवीकरणीय ऊर्जा	Fossil fuels, nuclear; finite reserves

One-Liner Revision -- Gases, Fuels and Energy Chemistry

#	Fast Fact
1	Calorific values (approx): Hydrogen = 142 MJ/kg; CNG/methane = 55 MJ/kg; Petrol = 47 MJ/kg
2	Highest calorific value fuel = **Hydrogen** (142 MJ/kg; "fuel of the future")
3	LPG composition: Propane (C3H8) + Butane (C4H10); stored as liquid; ethyl mercaptan = smell agent
4	CNG = mainly **methane (CH4)**; stored at 200-250 bar; cleanest fossil fuel
5	Coal types by carbon %: Peat (60%) < Lignite (70%) < Bituminous (80%) < Anthracite (90-95%)
6	Best quality coal = **Anthracite** (max carbon, max calorific value)
7	Petroleum (crude oil) = fractional distillation: gas > petrol > kerosene > diesel > fuel oil > bitumen
8	Octane number 87 = regular petrol; 91-93 = premium; higher = less knocking
9	Knocking in engine = premature ignition; prevented by anti-knock additives (earlier: TEL; now: MTBE)
10	Producer gas (CO + N2) = made by blowing air through burning coke
11	Water gas (CO + H2) = made by steam over hot coke; also called "synthesis gas"
12	Biogas = mainly methane; from cow dung + water; anaerobic digestion
13	Hydrogen fuel cell = H2 + O2 -> H2O + electricity; zero emission vehicle
14	Ozone (O3) layer = 15-35 km altitude (stratosphere); absorbs UV-B radiation
15	CFCs (chlorofluorocarbons) = destroy ozone; one Cl atom destroys 100,000 ozone molecules
16	Montreal Protocol (1987) = international treaty to phase out ozone-depleting substances
17	CO2 in atmosphere = ~420 ppm (2024); pre-industrial = ~280 ppm; rising due to fossil fuels
18	Methane GWP = **25x CO2** (100-year); N2O = 298x CO2; SF6 = 22,800x CO2
19	Solar panels = photovoltaic cells; convert sunlight to DC electricity; silicon-based
20	Wind energy = kinetic energy of wind -> electricity via turbine generator
21	Geothermal energy = heat from Earth's interior; Iceland, Philippines, India (Puga Valley, Ladakh)
22	Tidal energy = from rise and fall of ocean tides; predictable; coastal locations
23	Nuclear energy: fission (U-235, Pu-239) = current reactors; fusion = future
24	Hydroelectric power = 3rd largest renewable in India; HP, Uttarakhand, NE India
25	India's largest solar project = Bhadla Solar Park (Rajasthan; 2.2 GW)
26	India's target: 500 GW renewable energy by 2030 (Paris Agreement commitment)
27	Ethanol blending in India: E20 target = 20% ethanol in petrol by 2025
28	Hydrogen Mission (2021) = Green Hydrogen from electrolysis using renewable electricity

#	Fast Fact
29	Coal India Ltd (CIL) = world's largest coal mining company; state-owned
30	India's coal reserves = mainly in Jharkhand, Odisha, Chhattisgarh, WB